

OSKALOOSA, IA, USA

WMO: 720351

Lat: 41.226N

Lon: 92.491W

Elev: 256

StdP: 98.29

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 54919

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.3	-17.5	-24.0	0.4	-20.0	-21.5	0.6	-17.0	14.3	-3.8	12.8	-4.7	3.9	320	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	33.3	24.0	32.2	23.6	31.0	22.8	26.1	31.3	25.0	29.9	24.0	28.9	4.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.7	20.4	29.9	23.1	18.4	28.0	22.5	17.8	27.3	82.4	31.6	77.4	29.9	73.3	29.0	28.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.0	10.7	9.2	-25.3	36.1	4.6	1.8	-28.6	37.4	-31.3	38.5	-33.9	39.5	-37.3	40.9		
			-24.9	27.6	3.9	0.8	-27.7	28.2	-29.9	28.7	-32.1	29.1	-34.9	29.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	-4.4	-4.4	4.4	10.8	17.3	22.5	23.9	22.4	19.0	11.8	4.5	-1.4	(6)
	DBStd	11.46	6.91	6.56	6.53	5.53	4.83	3.05	3.32	2.96	4.37	5.29	5.83	6.23	(7)
	HDD10.0	1688	447	404	197	57	4	0	0	0	1	41	181	356	(8)
	HDD18.3	3385	706	637	434	233	79	4	2	3	45	212	416	613	(9)
	CDD10.0	1905	0	0	23	79	229	375	431	384	270	97	14	1	(10)
	CDD18.3	561	0	0	2	6	46	129	174	129	64	10	0	0	(11)
	CDH23.3	5338	0	0	14	98	439	1186	1760	1129	604	107	1	0	(12)
	CDH26.7	1923	0	0	1	24	134	431	723	379	205	25	0	0	(13)
(14) Wind	WSAvg	4.1	5.1	4.9	5.0	5.4	4.7	3.8	2.7	2.3	3.0	3.9	4.7	4.5	(14)
(15) Precipitation	PrecAvg	928	31	34	57	93	119	149	95	117	78	65	53	37	(15)
	PrecMax	1341	64	92	119	212	279	314	281	286	225	164	149	83	(16)
	PrecMin	623	7	6	8	24	12	30	10	24	22	11	4	3	(17)
	PrecStd	201	15	21	34	39	58	72	72	73	45	42	34	22	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	15.3	25.6	30.0	32.9	33.8	36.7	34.1	33.3	30.1	22.7	15.6	(19)
		MCWB	7.4	9.8	17.2	18.1	21.8	23.3	24.4	23.8	23.0	20.6	14.5	11.8	(20)
	2%	DB	9.3	11.1	21.0	26.2	30.2	32.5	34.0	32.3	31.2	26.6	19.1	11.8	(21)
		MCWB	5.9	7.1	15.0	16.5	19.7	23.0	24.9	24.5	22.0	18.3	13.5	8.7	(22)
	5%	DB	6.5	7.0	17.5	23.0	27.7	31.2	32.3	30.7	29.1	23.7	16.0	9.1	(23)
		MCWB	4.1	4.2	13.0	14.5	18.6	22.1	24.7	23.5	21.3	17.0	10.9	7.1	(24)
	10%	DB	4.1	3.8	14.3	20.6	25.9	29.4	30.9	28.8	27.2	21.1	12.9	6.5	(25)
		MCWB	2.2	2.0	10.1	13.1	18.1	21.4	23.8	22.5	19.7	15.3	9.1	4.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.8	10.8	18.1	19.7	23.0	26.3	27.3	27.3	24.7	22.8	16.4	13.7	(27)
		MCDB	11.6	13.8	23.8	26.8	30.1	31.5	33.0	32.3	30.4	28.0	20.5	14.8	(28)
	2%	WB	6.2	7.2	16.0	17.8	21.7	24.8	26.3	25.8	23.5	20.4	14.1	9.4	(29)
		MCDB	9.0	10.0	19.6	23.6	28.0	29.8	32.1	30.2	29.1	23.9	17.8	10.7	(30)
	5%	WB	4.3	4.1	13.7	15.8	20.6	23.7	25.3	24.6	22.5	18.3	11.8	7.1	(31)
		MCDB	6.7	6.7	17.0	21.4	26.0	29.0	30.7	28.8	27.3	21.6	15.1	9.0	(32)
	10%	WB	2.4	2.1	10.4	14.4	19.4	22.7	24.3	23.4	21.1	16.3	9.6	4.5	(33)
		MCDB	4.1	3.7	14.1	19.5	24.0	27.8	29.1	27.6	25.0	20.4	12.9	6.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	9.2	10.1	12.1	11.4	11.1	11.4	11.4	12.7	11.9	10.6	8.6	(35)
		MCDBR	11.6	12.1	15.0	16.9	13.9	13.0	12.5	12.8	14.0	15.0	14.6	11.5	(36)
	5% WB	MCWBR	8.7	8.7	9.1	8.9	6.1	5.8	5.6	6.4	6.0	7.6	9.2	8.8	(37)
		MCWBR	11.3	12.0	12.9	14.4	11.7	11.0	11.3	10.7	11.7	11.8	13.5	10.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.291	0.307	0.335	0.385	0.413	0.421	0.438	0.419	0.386	0.341	0.299	0.289	(40)
		taud	2.419	2.372	2.388	2.262	2.254	2.280	2.218	2.290	2.332	2.458	2.503	2.441	(41)
	Ebn at Noon		867	903	915	885	865	856	837	843	848	855	854	837	(42)
		Edh at Noon	79	97	107	131	135	132	139	126	112	88	73	72	(43)
(44) All-Sky Solar Radiation	RadAvg		1.90	2.90	3.78	4.75	5.56	6.17	6.21	5.42	4.64	3.09	2.09	1.60	(44)
	RadStd		0.19	0.22	0.34	0.35	0.40	0.63	0.45	0.32	0.34	0.35	0.18	0.15	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page